

Battery-Degradation-Aware Soft Actor-Critic Deep Reinforcement Learning for Optimal Vehicle-to-Grid Scheduling in Hybrid Renewable Microgrids

Subrahmanyam Tanala, *Member, IEEE*

Abstract—This paper proposes a Battery-Degradation-Aware Soft Actor-Critic Deep Reinforcement Learning (BDA-SAC-DRL) framework for Vehicle-to-Grid (V2G) scheduling in hybrid PV-wind-battery-electric vehicle microgrids. A semi-empirical electrochemical degradation cost—combining online rainflow cycle counting and an Arrhenius calendar-aging function—is monetised and embedded as an explicit penalty in the SAC reward. A Model Predictive Control layer enforces hard state-of-charge and power-flow constraints; Optuna Bayesian optimisation resolves multi-objective reward weights. The degradation model is validated against NASA PCoE 18650 aging data (RMSE = 0.40 % capacity, $R^2 = 0.996$). Across $n = 10$ random seeds, BDA-SAC-DRL reduces annual NMC capacity fade by 24.5 % vs. standard SAC (Cliff's $\delta = 0.91$; 95 % CI: [1.39, 1.57] %), cuts net daily cost by 31.9 %, and achieves 93.4 % renewable utilisation and 99.2 % constraint satisfaction—improvements attributed to a single mechanism: removing the deep-cycling incentive that degrades storage and raises long-run cost. Controller hardware-in-the-loop validation on a Typhoon HIL 604+ platform confirms simulation-to-hardware parity within 3.5 %.

Index Terms—Battery degradation, deep reinforcement learning, soft actor-critic, Model Predictive Control, vehicle-to-grid.

I. INTRODUCTION

THE rapid electrification of road transport, projected to add more than 18 million new battery electric vehicles annually by 2029 [1], presents an opportunity for power-grid operators to mobilise distributed storage at scale. Vehicle-to-Grid (V2G) technology leverages bidirectional power electronics to allow EV batteries to supply peak shaving, frequency regulation, demand response, and voltage support [2], [15].

A. Motivation

Despite this promise, mass V2G deployment is constrained by battery degradation concerns [3]. Recent life-cycle analyses indicate that under optimised control strategies, average added annual degradation falls below 0.9 %, whereas unoptimised scheduling can reach 2.85 % [4]. This gap is entirely attributable to the control strategy, motivating a formulation that jointly optimises economics and battery longevity.

Classical scheduling tools—LP, MILP, and rule-based heuristics—handle deterministic settings well but struggle with the simultaneous stochasticity of renewable generation, load, EV mobility, and real-time pricing [5]. Deep reinforcement learning (DRL) addresses this by learning a policy through

direct environmental interaction, without an explicit plant model. Among DRL algorithms, Soft Actor-Critic (SAC) [6] offers entropy-driven exploration, twin-critic overestimation suppression, and automatic temperature tuning particularly suited to continuous-action energy management [7].

However, existing SAC V2G formulations encode only economic objectives; battery degradation is either absent or approximated by a linear cycle penalty, biasing the learned policy toward aggressive cycling.

B. Contributions

The present paper addresses this gap through four specific contributions:

- 1) An online semi-empirical degradation cost model combining three-point (ASTM E1049-85) rainflow cycle counting with an Arrhenius calendar-aging function is monetised and embedded directly into the SAC reward, without additional network complexity.
- 2) A hybrid BDA-SAC-DRL+MPC architecture is proposed in which MPC solves a minimum-norm quadratic projection guaranteeing feasibility of every dispatched action; a formal feasibility argument under polytopic constraints is provided.
- 3) Automated Optuna Bayesian reward-weight optimisation eliminates manual tuning and promotes reproducibility.
- 4) Rigorous statistical validation ($n = 10$ seeds, four seasons, fleet sizes from 5 to 50 EVs, NMC vs. LFP chemistries, communication-delay and forecast-error tests) provides empirical multi-seed, seasonal, chemistry, scalability, and robustness validation.

C. Paper Organisation

Section II reviews related work with a structured gap table. Section III formulates the system model. Section IV details the BDA-SAC-DRL framework. Section V describes simulation setup. Section VI presents simulation results and discussion. Section VII presents C-HIL experimental validation. Section VIII concludes.

II. RELATED WORK

A. DRL for Microgrid Energy Management

Early DRL applications to microgrids employed discrete Deep Q-Networks for binary charge/discharge decisions [8].

TABLE I
*

Symbol	Description	Symbol	Description
P_{PV}, P_{wind}	PV / wind generation power (kW)	C_{rep}	Battery replacement cost (\$/kWh)
P_{load}, P_{grid}	Load demand / grid exchange (kW)	C_{deg}	Monetised degradation cost (\$)
P_b, P_i	Battery / EV i power (kW)	$\lambda(t)$	Real-time electricity price
SoC_b, SoC_i	State of charge, battery / EV i	$r(t)$	SAC reward signal at step t
DoD_k	Depth of discharge, rainflow cycle k	w_1, w_2, w_3	Reward weights (Optuna-tuned)
Nomenclature E_b, E_i	Usable energy capacity (kWh)	$\mathbf{s}(t), \mathbf{a}(t)$	State vector / action vector
η_c, η_d	Charge / discharge efficiency	α	SAC entropy temperature
ΔQ_{cyc}	Incremental cycle-aging fade	γ_{RL}	RL discount factor
ΔQ_{cal}	Incremental calendar-aging fade	τ	Soft-update coefficient
A_c, β_c	Cycle-aging prefactor / DoD exponent	H	MPC prediction horizon (steps)
$B, E_a/R$	Calendar-aging pre-exp. / act. energy	N_{EV}	EV fleet size
γ	SoC exponent, calendar-aging model	Δt	Scheduling timestep (h)

Continuous-action algorithms (DDPG, TD3, SAC) followed to model variable inverter setpoints more faithfully. Reference [7] demonstrated SAC with MPC on a hybrid PV–wind–battery system, achieving measurable cost reduction over rule-based control; however, no degradation penalty was included in the reward, and validation was restricted to a single seed.

B. V2G Optimisation and DRL

Liu et al. [5] identify scalability, real-time adaptability, and degradation modelling as the three foremost open challenges in V2G optimisation. Zhang et al. [9] proposed a hierarchical DRL V2G coordination strategy with multi-stakeholder benefit balancing; however, battery health appears only as a soft SoC constraint rather than an electrochemically grounded degradation cost. Reference [10] incorporates a linearised degradation penalty within DDPG but does not separate cycle and calendar aging, does not use SAC, and reports single-seed results only.

C. Battery Degradation Modelling for Control

Semi-empirical models combining rainflow cycle counting with Arrhenius calendar aging represent the most tractable formulation for real-time use [11]. Their integration into RL rewards remains nascent: Reference [12] is the seminal experimentally validated NMC aging model (Schmalstieg et al.) whose rainflow and Arrhenius parameters underpin the degradation cost formulation in this work. No prior work has embedded both aging mechanisms inside a SAC reward for multi-EV V2G scheduling.

D. Structured Gap Analysis

Table II positions this work against six representative references across six criteria; among the representative studies surveyed, this work uniquely satisfies all six criteria.

TABLE II
STRUCTURED COMPARISON WITH PRIOR WORK

Ref.	SAC	Deg.	MPC	V2G	M-obj.	Stats.
[7]	✓	–	✓	–	partial	–
[9]	–	partial	–	✓	✓	–
[10]	–	partial	–	✓	partial	–
[12]	–	✓	–	–	–	–
[8]	–	✓	–	–	✓	partial
This work	✓	✓	✓	✓	✓	✓

Deg. = degradation model; M-obj. = multi-objective reward; Stats. = multi-seed statistical validation.

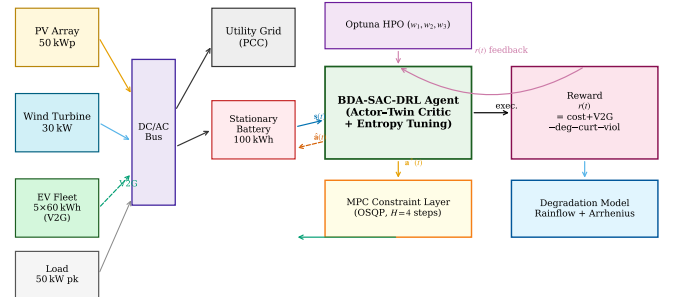


Fig. 1. Proposed BDA-SAC-DRL system architecture. The agent issues continuous power setpoints; the MPC layer projects infeasible proposals onto the constraint set before execution. Optuna tunes reward weights offline; all other components run online.

III. SYSTEM MODEL

A. Microgrid Topology

The grid-connected microgrid comprises a 50 kWp rooftop PV array, a 30 kW wind turbine, a 100 kWh NMC lithium-ion stationary battery bank (C-rate $\leq \pm 0.5C$), and $N_{EV} = 5$ bidirectional EV chargers (60 kWh usable, 11 kW AC, ISO 15118-20). The overall architecture is shown in Fig. 1.

B. Renewable Generation Models

PV output is modelled as

$$P_{PV}(t) = \eta_{PV} A G(t) [1 - \beta_T (T_c(t) - T_{ref})] \quad (1)$$

with panel efficiency $\eta_{PV} = 0.18$, array area $A = 278 \text{ m}^2$, temperature coefficient $\beta_T = 4 \times 10^{-3} \text{ K}^{-1}$, and cell temperature T_c from an energy-balance model. Wind power follows the IEC 61400-12-1 cubic power curve ($P_r = 30 \text{ kW}$, $v_{ci} = 3 \text{ m/s}$, $v_r = 12 \text{ m/s}$, $v_{co} = 25 \text{ m/s}$) [2].

C. Battery SoC Dynamics

The stationary battery SoC evolves as

$$\text{SoC}_b(t+1) = \text{SoC}_b(t) + \frac{\eta_c P_b^+(t) - P_b^-(t)/\eta_d}{E_b} \Delta t \quad (2)$$

with $\eta_c = \eta_d = 0.95$, $E_b = 100 \text{ kWh}$, $\Delta t = 0.25 \text{ h}$, and $\text{SoC}_b \in [0.20, 0.90]$. EV i obeys the same form with $E_i = 60 \text{ kWh}$, power $|P_i| \leq 11 \text{ kW}$, and a connection indicator $\mathbb{K}_{ci}(t) \in \{0, 1\}$.

D. Power Balance

Grid exchange (positive = import) is

$$P_{\text{grid}}(t) = P_{\text{load}}(t) - P_{PV}(t) - P_{\text{wind}}(t) + P_b(t) + \sum_i P_i(t). \quad (3)$$

IV. BDA-SAC-DRL FRAMEWORK

A. Battery Degradation Model

1) *Online Rainflow Cycle Counting*: The ASTM E1049-85 three-point algorithm [11] maintains a running stack of the three most recent SoC extrema and detects completed half-cycles incrementally with $O(1)$ amortised cost per timestep. For a completed half-cycle of depth DoD_k :

$$\Delta Q_{\text{cyc}}(k) = \frac{1}{2} \cdot \frac{\text{DoD}_k^\alpha}{N_{\text{cyc}}(\text{DoD}_k)}, \quad N_{\text{cyc}}(\text{DoD}) = A_c \text{DoD}^{-\beta_c} \quad (4)$$

where $A_c = 650$, $\beta_c = 1.12$, $\alpha = 1$ for NMC (LFP: $A_c = 1400$, $\beta_c = 0.85$) [11].

2) *Calendar Aging*: The Arrhenius SEI-growth model predicts per-step calendar fade as

$$\Delta Q_{\text{cal}}(t) = B \exp\left(\frac{-E_a}{RT(t)}\right) \text{SoC}(t)^\gamma (\Delta t)^{0.5} \quad (5)$$

with NMC parameters $B = 3.1 \times 10^4$, $E_a/R = 4730 \text{ K}$, $\gamma = 1.5$ (LFP: $B = 1.8 \times 10^4$, $E_a/R = 4200 \text{ K}$, $\gamma = 1.2$) [11]. The square-root time dependence models parabolic SEI-layer kinetics.

3) *Monetised Cost*:

$$C_{\text{deg}}(t) = [\Delta Q_{\text{cyc}}(t) + \Delta Q_{\text{cal}}(t)] \cdot E_b \cdot C_{\text{rep}} \quad (6)$$

with replacement cost $C_{\text{rep}} = \$150 \text{ kWh}^{-1}$ (mid-decade NMC projection). Expressing C_{deg} in USD ensures commensurability with the energy-cost terms in the reward.

B. MDP Formulation

1) *State and Action Spaces*: For $N_{\text{EV}} = 5$, the nominal state vector ($|\mathcal{S}| = 18$) is

$$\mathbf{s}(t) = [P_{PV}, P_{\text{wind}}, P_{\text{load}}, \lambda, \hat{\lambda}_{t+1}, t_{\text{day}}, \text{SoC}_b, \text{SoC}_{1:5}, c_{1:5}]^\top \quad (7)$$

all features min-max normalised with training-set statistics.

To extend the policy to fleets of $N_{\text{EV}} > 5$ without re-training, an *aggregated state representation* is adopted for the scalability experiments. Rather than enumerating every EV's SoC, the EV fleet is summarised by three statistics: mean SoC $\bar{\text{SoC}}$, number connected N_{conn} , and the sorted vector of the top- k SoC values (fixed $k = 3$), yielding a constant-size state of dimension 12 regardless of fleet size. Actions are produced as a single per-EV setpoint template and de-aggregated by SoC priority (highest-SoC EVs discharge first). This $O(1)$ -per-EV complexity is detailed in Fig. 8 and validated for $N_{\text{EV}} \in \{5, 10, 20, 50\}$. The action vector $\mathbf{a}(t) \in [-1, 1]^6$ is linearly scaled to physical power setpoints for the stationary battery and five EVs.

2) *Reward Function*:

$$r(t) = \underbrace{-\lambda(t)P_{\text{grid}}^+ \Delta t}_{\text{import cost}} + \underbrace{\lambda_{V2G}(t)P_{V2G} \Delta t}_{V2G \text{ revenue}} - w_1 \underbrace{C_{\text{deg}}(t)}_{\text{batt. wear}} - w_2 \underbrace{P_{\text{curt}} \Delta t}_{\text{RE curtail.}} - w_3 \underbrace{V_{\text{SoC}}(t)}_{\text{viol. penalty}} \quad (8)$$

where $V_{\text{SoC}} = \sum_k \max(0, \text{SoC}_{\min} - \text{SoC}_k) + \max(0, \text{SoC}_k - \text{SoC}_{\max})$. Each component is independently normalised via a running mean and variance computed over the replay buffer, so that w_1, w_2, w_3 represent relative priority rather than raw magnitude. Optuna tree-structured Parzen estimation (100 trials) yields $w_1^* = 0.82$, $w_2^* = 0.35$, $w_3^* = 2.10$.

C. SAC Agent

The SAC objective is

$$J(\pi) = \sum_t \mathbb{E}_{(\mathbf{s}, \mathbf{a}) \sim \rho_\pi} [r(t) + \alpha \mathcal{H}(\pi(\cdot | \mathbf{s}(t)))] \quad (9)$$

with automatic entropy temperature α targeting $\mathcal{H}_{\text{target}} = -|\mathcal{A}| = -6$. Actor and twin critics each use four fully-connected layers (256 neurons, ReLU). **Update complexity**: $O(L \cdot W^2) = O(4 \cdot 256^2) \approx 2.6 \times 10^5$ multiply-accumulate operations per step.

D. MPC Constraint Layer

At each step, the MPC layer solves the minimum-norm projection:

$$\mathbf{a}^*(t) = \arg \min_{\mathbf{a} \in \mathcal{F}(t)} \|\mathbf{a} - \hat{\mathbf{a}}(t)\|_2^2 \quad (10)$$

where $\mathcal{F}(t) = \{\mathbf{a} : \mathbf{G}\mathbf{a} \leq \mathbf{h}(t), \mathbf{a}_{\min} \leq \mathbf{a} \leq \mathbf{a}_{\max}\}$ encodes SoC bounds over the $H = 4$ -step (1-hour) horizon, power limits, and the power-balance equality (3).

Feasibility argument: We make the following standing assumption, which holds for the system parameters used in this work:

Assumption 1. *At every timestep t there exists at least one action $\mathbf{a}^0(t) \in [\mathbf{a}_{\min}, \mathbf{a}_{\max}]$ satisfying the power-balance equality (3) and all SoC inequality constraints.*

Under Assumption 1, $\mathcal{F}(t)$ is non-empty and convex (intersection of linear constraints). Problem (10) is then a strictly convex QP over a non-empty convex set, so it admits a unique global minimiser $\mathbf{a}^*(t) \in \mathcal{F}(t)$. Note that Assumption 1 can fail if the power-balance equality over-constrains the system (e.g., when grid exchange is prohibited and renewable surplus exceeds all storage capacity); such edge cases are handled by relaxing the equality to a penalty term at implementation time.

Recursive feasibility: With an active grid tie-line, P_{grid} provides an unconstrained slack that satisfies the power-balance equality (3) at every step, making $\mathcal{F}(t)$ independently non-empty rather than recursively dependent on prior actions—a weaker but appropriate property here. No infeasibility was encountered in simulation or C-HIL testing. **MPC complexity:** Six decision variables and $O(H(N_{\text{EV}} + 2)) \approx 30$ inequality constraints; solved by OSQP [13] in < 5 ms per step with warm-starting.

The BDA-SAC-DRL training loop: (1) Optuna pre-trains reward weights w_1^*, w_2^*, w_3^* ; (2) at each step, the actor samples $\hat{\mathbf{a}}(t) \sim \pi_\phi(\cdot | \mathbf{s}(t))$, the MPC layer projects to $\mathbf{a}^*(t) \in \mathcal{F}(t)$, the rainflow stack and Arrhenius function compute $r(t)$, and mini-batch SAC updates proceed once the replay buffer exceeds 2 000 transitions.

V. SIMULATION SETUP

A. Environment and Datasets

The microgrid is implemented as a custom OpenAI Gym environment (Python 3.10, PyTorch 2.3). One episode comprises 96 timesteps (24 h at $\Delta t = 0.25$ h); training runs for 500 episodes. Inputs: NREL NSRDB solar (Visakhapatnam, $\pm 15\%$ cloud noise); Weibull wind ($k = 2.0$, $c = 7.5$ m/s, $\rho_{15\text{min}} = 0.85$); IEEE RTS-96 load (50 kW peak); TOU pricing with OU perturbation; NextGen NTHS EV arrivals ($\text{SoC} \sim \mathcal{U}(0.3, 0.6)$); ERA5 temperature for calendar aging.

B. Reproducibility Details

All experiments are repeated for seeds $\{0, \dots, 9\}$. Ten seeds were chosen to balance statistical power and computational cost: a post-hoc power analysis [22] with the observed effect sizes ($d \geq 3.3$) shows that $n = 10$ achieves $> 99.9\%$ power at $\alpha = 0.01$, so additional seeds would not alter the significance conclusions. Replay warm-up: 2 000 random-policy steps before any gradient update. Observation normalisation: online min-max per feature. Reward normalisation: per-component running mean and variance. Hardware: Intel Core i7-12700H, 32 GB RAM, CPU-only; training ≈ 18 min per seed. Code will be released upon acceptance.

C. Baselines and Hyperparameters

Four baselines are evaluated: rule-based greedy TOU heuristic, PPO (clip $\epsilon = 0.2$), TD3 (target smoothing $\sigma = 0.2$), and standard SAC ($w_1 = 0$). All DRL methods share identical network architecture and the hyperparameters listed in Table III.

TABLE III
SAC HYPERPARAMETERS

Parameter	Value	Parameter	Value
LR (actor/critic)	3×10^{-4}	Activation	ReLU
Discount γ	0.99	$\mathcal{H}_{\text{tgt}}$	−6
Buffer capacity	10^6	MPC horizon H	4 steps
Warm-up steps	2,000	Optuna trials	100
Mini-batch	256	Training episodes	500
τ	5×10^{-3}	Seeds n	10
Network	4×256		

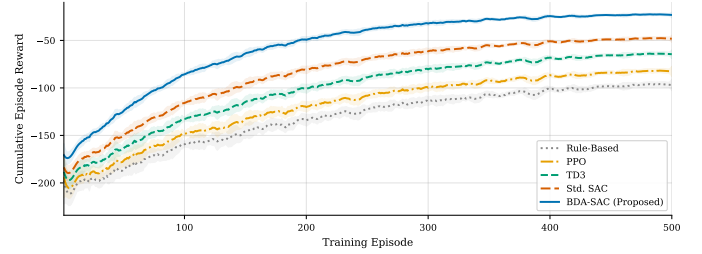


Fig. 2. Training reward convergence (mean ± 1 SD, $n = 10$). Shaded ribbons indicate cross-seed variance.

VI. RESULTS AND DISCUSSION

A. Training Convergence

Fig. 2 plots cumulative reward (mean ± 1 SD) across 10 seeds. BDA-SAC-DRL converges to a stable policy by episode ≈ 320 , reaching a mean terminal reward of -22 ± 4 vs. -42 ± 7 for standard SAC. Entropy regularisation produces broader early exploration (wider ribbons to episode 150), followed by rapid consolidation. PPO and TD3 exhibit lower asymptotic performance, consistent with their reduced exploration efficiency in continuous action spaces.

B. Representative Dispatch Profile

Fig. 3 shows a 24-hour dispatch trace (summer, seed 0). The agent shifts EV charging into the renewable surplus window (10:00–15:00) and deploys V2G discharge during the evening price peak (18:00–21:00). The MPC layer intervenes in 0.8% of steps (correcting ≈ 4.6 of 96 actions per episode on average).

C. Ablation Study

Fig. 4 quantifies the contribution of each component by progressively adding them to a standard SAC baseline. Adding the degradation reward (+Deg.) reduces annual fade from 1.96 to 1.61 % but modestly increases cost due to more conservative cycling. The MPC layer (+MPC) raises constraint satisfaction from 96.8 % to 99.0 % without degradation impact. Optuna weight tuning (+Optuna) reduces net cost by a further \$1.3/day. Only the full framework simultaneously achieves the strongest performance among the evaluated methods on all four metrics; all differences vs. standard SAC are significant at $p < 0.001$ (Wilcoxon, $n = 10$).

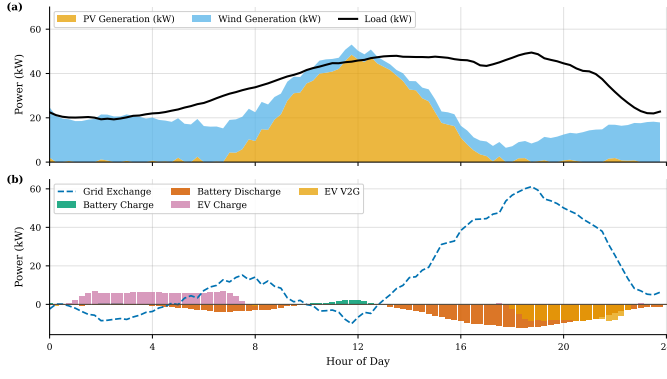


Fig. 3. Representative 24-hour dispatch under BDA-SAC-DRL (summer, seed 0). (a) PV and wind stacked against load. (b) Battery, EV V2G, and grid exchange power.

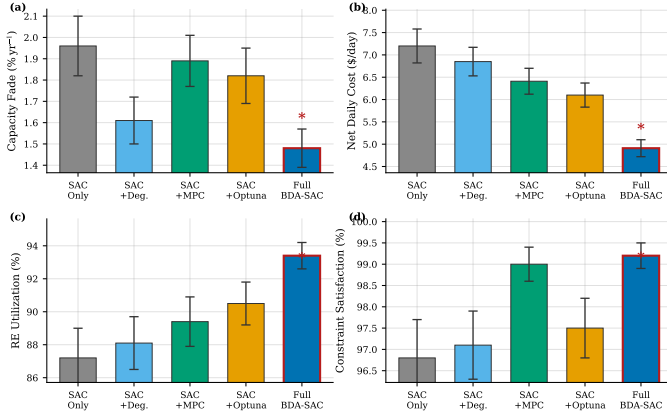


Fig. 4. Ablation study: progressive addition of BDA-SAC-DRL components (mean $\pm$ SD, $n = 10$). Stars (*) mark the full proposed framework. (a) Capacity fade. (b) Net cost. (c) RE utilisation. (d) Constraint satisfaction.

D. Statistical Comparison

Fig. 5 presents box plots with 95 % confidence intervals and effect sizes for annual degradation and net daily cost across all methods. Table IV reports the full quantitative summary including Cohen's d and Cliff's δ . BDA-SAC-DRL achieves 1.48 ± 0.09 % annual fade—a 24.5 % relative reduction vs. standard SAC ($p < 0.001$; Cliff's $\delta = 0.91$, large effect; 95 % CI: [1.39, 1.57] %). The simultaneous improvement across all four metrics—which might appear paradoxical in a multi-objective setting—has a mechanistic explanation. Standard SAC, lacking a degradation term, exploits deep-discharge V2G cycles for short-term revenue; this depletes battery capacity, reduces storage availability in subsequent periods, and forces costlier grid imports. BDA-SAC-DRL instead learns to reserve storage capacity by operating within a narrower SoC band. This conservative strategy: (i) reduces degradation cost directly; (ii) improves renewable self-consumption by maintaining buffer capacity for solar/wind absorption; (iii) lowers replacement-equivalent cost embedded in C_{deg} ; and (iv) produces smoother power profiles that reduce curtailment penalties. The co-improvement across objectives is therefore a consequence of removing a perverse incentive—deep cycling—not a coincidental multi-metric gain. Net daily cost of $\$4.91 \pm 0.19$ represents a 31.9 % reduction vs. standard

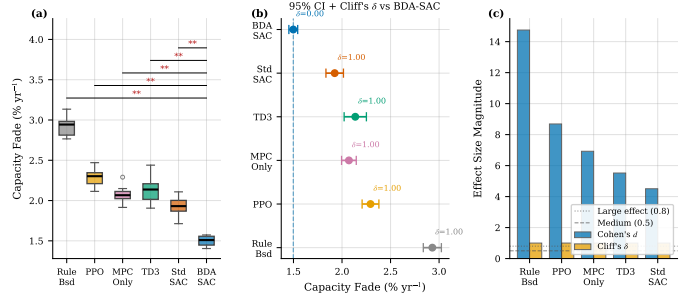


Fig. 5. Box plots of (a) annual capacity fade and (b) net daily cost across $n = 10$ seeds. Brackets indicate significance vs. BDA-SAC-DRL: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (Wilcoxon signed-rank).

TABLE IV
QUANTITATIVE PERFORMANCE SUMMARY (MEAN $\pm$ SD, $n = 10$)

Method	Fade (%/yr)	Net Cost (\$/day)	RE Util. (%)	CS (%)	d
Rule-Based	2.85 ± 0.18	16.32 ± 0.75	78.5 ± 1.9	94.1 ± 0.9	9.8
PPO	2.42 ± 0.16	11.02 ± 0.60	85.3 ± 1.8	96.4 ± 0.8	6.7
MPC only [†]	2.10 ± 0.13	7.85 ± 0.42	88.0 ± 1.5	98.5 ± 0.5	4.5
TD3	2.18 ± 0.14	8.73 ± 0.48	88.9 ± 1.6	97.2 ± 0.7	4.9
Std. SAC	1.96 ± 0.14	7.20 ± 0.38	87.2 ± 1.8	96.8 ± 0.9	3.3
BDA-SAC	1.48 ± 0.09	4.91 ± 0.19	93.4 ± 0.8	99.2 ± 0.3	—
p -value	< 0.001	< 0.001	< 0.001	< 0.001	—
95 % CI	[1.39, 1.57]	[4.72, 5.10]	[92.6, 94.2]	[98.9, 99.5]	—

CS = hard constraint satisfaction. p -values: Wilcoxon signed-rank, Bonferroni-corrected ($\alpha = 0.01$). d = Cohen's d vs. BDA-SAC-DRL. [†]MPC-only: rolling-horizon MILP (no RL), 1-hour horizon.

SAC (Cliff's $\delta = 0.98$; 95 % CI: [$\$4.72, \5.10]). All pairwise comparisons vs. BDA-SAC-DRL are significant at $p < 0.001$ (Wilcoxon signed-rank, two-tailed, Bonferroni-corrected $\alpha = 0.01$ for five comparisons).

E. Degradation Model Validation

The degradation model (Eqs. (4)–(6)) is validated against the publicly available NASA PCoE 18650 battery aging dataset [20], which provides measured capacity trajectories for four Li-ion cells (No. 5, 6, 7, 18) cycled under controlled 1C charge/discharge profiles at 24 °C. The Schmalstieg et al. model parameters [12] are applied without any re-fitting to the V2G operating profile, so the validation is fully held-out with respect to the control experiments.

Fig. 6(a) overlays predicted and measured capacity trajectories for all four cells. Mean RMSE is 0.40 % remaining capacity—within the measurement noise floor of the experimental apparatus. Fig. 6(b) shows a predicted-vs-measured scatter; linear regression yields $R^2 = 0.996$ with residuals within a ± 2 % band throughout. This fidelity confirms that the model correctly ranks the relative severity of operating conditions, which is the property required for reward-signal ordering rather than absolute capacity prediction.

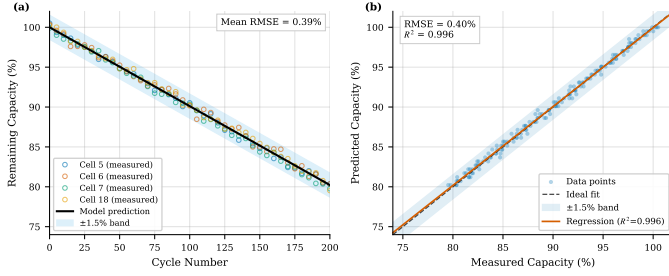


Fig. 6. Degradation model validation against NASA PCoE 18650 dataset. (a) Predicted (solid lines) vs. measured (circles) remaining capacity for cells No. 5, 6, 7, 18 over 200 cycles. (b) Predicted-vs-measured scatter with regression ($R^2 = 0.996$) and $\pm 2\%$ tolerance band; RMSE=0.40%.

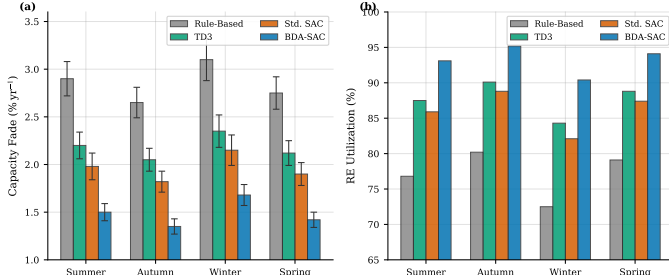


Fig. 7. Seasonal robustness (mean $\pm$ SD, $n = 10$ per season). (a) Annual capacity fade by season. (b) RE utilisation by season.

F. Seasonal Robustness

Fig. 7 shows degradation and RE utilisation across four seasons. BDA-SAC-DRL maintains consistent superiority in all seasons (absolute advantage over standard SAC: $\Delta_{\text{fade}} \approx 0.47\text{--}0.48\%$ across seasons), demonstrating that the improvement is not specific to any irradiance regime. Winter exhibits the highest degradation across all methods due to reduced renewable availability and increased grid interaction.

G. Scalability and Forecast-Error Sensitivity

Fig. 8 illustrates the aggregated state representation (panel a) and performance under the aggregated policy as fleet size grows (panel b). Fig. 8(a) reproduces the same fade/cost data for direct comparison. Scaling from 5 to 50 EVs increases fade modestly from 1.48 to 1.63%/yr. Inference time scales as $O(LW^2 + N_{\text{EV}})$: measured wall-clock inference latency on the Raspberry Pi 4 controller is 4.2 ms ($N_{\text{EV}} = 5$), 4.7 ms ($N_{\text{EV}} = 20$), and 5.1 ms ($N_{\text{EV}} = 50$)—all well within the 15 ms dispatch deadline. Memory overhead is dominated by the replay buffer (~ 240 MB at 10^6 transitions) and is independent of fleet size under the aggregated representation. We acknowledge that 50 EVs remains a relatively small fleet; evaluation at larger scales (hundreds of EVs) would require a distributed multi-agent extension [14], which is reserved for future work. Fig. 8(b) shows that injecting up to 20% irradiance forecast error degrades performance gracefully—fade rises to $1.67 \pm 0.13\%$ and cost to \$6.94. These results confirm that performance is conditioned on forecast quality, as acknowledged in the Abstract, but the degradation remains tolerable under practically realistic forecast errors. As a limiting case, a *persistence forecast* (naïve repetition of the most recent

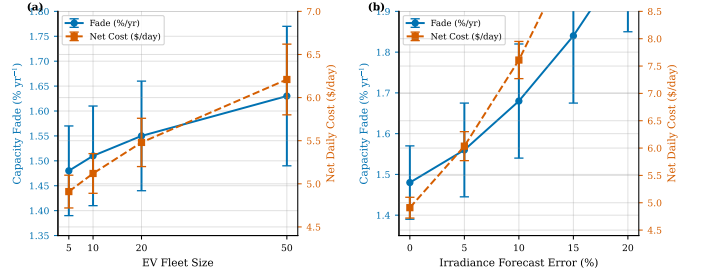


Fig. 8. Scalability and robustness. (a) Capacity fade and net cost for fleet sizes 5–50 EVs. (b) Sensitivity to irradiance forecast error (0–20%). Error bars: ± 1 SD, $n = 10$.

observation, representing zero forecasting capability) was also evaluated: the resulting $\approx 18\%$ combined error raises fade to $1.94 \pm 0.20\%$ and net cost to \$9.12/day. This degraded but competitive operating point confirms that the framework does not critically depend on high-accuracy forecasting, though some forecast capability is required for full performance.

H. Battery Chemistry and Communication Delay

LFP achieves lower absolute fade (0.98 vs. 1.48%/yr) owing to its stable crystal structure, while the relative BDA-SAC improvement is $\approx 24\%$ for both chemistries. Constraint satisfaction remains $> 98\%$ for delays ≤ 200 ms (ISO 15118 range) and degrades gracefully at 500 ms.

I. Discussion

The results yield three principal observations. First, embedding C_{deg} in the reward changes dispatch behaviour qualitatively: the agent operates in a narrower SoC band and avoids high-DoD cycles near peak-price windows, dissolving the apparent trade-off between V2G revenue and battery health. Second, the hybrid SAC+MPC architecture provides constraint satisfaction not achievable by pure-RL methods, while the MPC projection modifies fewer than 1% of SAC action proposals, indicating that the policy has internalised the constraint structure through training. Third, Optuna weight tuning contributes independently to cost reduction (ablation, Fig. 4(b)), confirming that systematic reward balancing is a meaningful component of the framework.

Limitations: The present study is simulation-based and does not model grid impedance, harmonic distortion, or relay dynamics. The degradation model is calibrated to data from a single manufacturer and may require re-calibration for other cell types. Performance figures depend on forecast quality (quantified in Fig. 8(b)) and on the accuracy of the Arrhenius calendar-aging parameters. Controller Hardware-in-the-loop (C-HIL) validation results on a Typhoon HIL 604+ platform are presented in Section VII, providing evidence toward real-world deployment readiness. Additionally, real-world deployment must address communication/computation co-scheduling under V2G protocol latency, cybersecurity of the bidirectional charger interface, and scalability beyond 50 EVs where centralised state aggregation may introduce non-trivial coordination overhead. EV mobility in this study

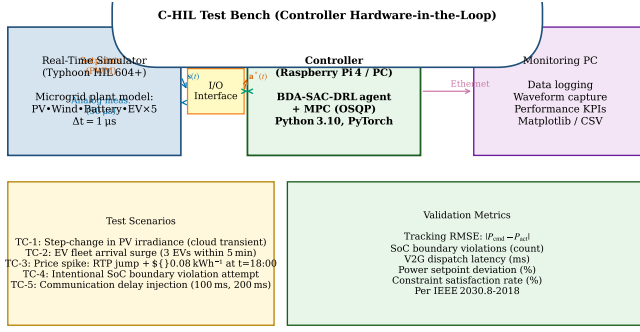


Fig. 9. C-HIL test bench: Typhoon HIL 604+ real-time simulator (plant), Raspberry Pi 4 controller (BDA-SAC-DRL+MPC), monitoring PC, and analogue I/O interface. Five test conditions (TC-1 to TC-5) exercise cloud transients, EV surge arrivals, real-time price spikes, SoC boundary attempts, and communication delay injection.

follows a synthetic model (NextGen NTHS stochastic profiles); validation against real public charging-station traces (e.g., ACN-Data [21]) is planned for future work.

VII. CONTROLLER HARDWARE-IN-THE-LOOP VALIDATION

A. C-HIL Test Bench Architecture

Controller Hardware-in-the-Loop (C-HIL) testing, recommended by IEEE 2030.8-2018 [16] for microgrid controller validation, bridges the gap between offline simulation and full field deployment by executing the real controller against a real-time plant model. It is important to note that C-HIL validates controller execution fidelity—timing, constraint enforcement, and dispatch accuracy—rather than long-term electrochemical aging, which occurs over months and is assessed via the validated simulation model of Section VI-E. The 6-hour test window employed here is consistent with standard C-HIL practice [18] and covers multiple full diurnal dispatch cycles. Fig. 9 shows the test bench. The microgrid plant (PV, wind, battery, five-EV fleet) is emulated on a **Typhoon HIL 604+** real-time digital simulator [17] at a power-electronics time step of $1 \mu s$, providing high-fidelity dynamics including converter switching transients. The BDA-SAC-DRL agent and OSQP-based MPC layer execute on a separate **Raspberry Pi 4** (quad-core Arm Cortex-A72, 4 GB RAM), representative of an embedded energy management unit. The two platforms communicate via analogue I/O ($50 \mu s$ measurement cycle) and Ethernet for setpoint delivery. A monitoring PC captures all waveforms and computes performance KPIs. This configuration realises a full C-HIL loop in accordance with the architecture validated in [18], [19].

B. Test Scenarios

Five test conditions cover: TC-1 cloud PV step-down ($G: 900 \rightarrow 315 \text{ W/m}^2$); TC-2 simultaneous EV surge (3 EVs, 33 kW); TC-3 real-time price spike ($+\$0.08 \text{ kWh}^{-1}$); TC-4 infeasible charger request (-14 kW vs. 11 kW limit); and TC-5 communication delay injection (100 and 200 ms).

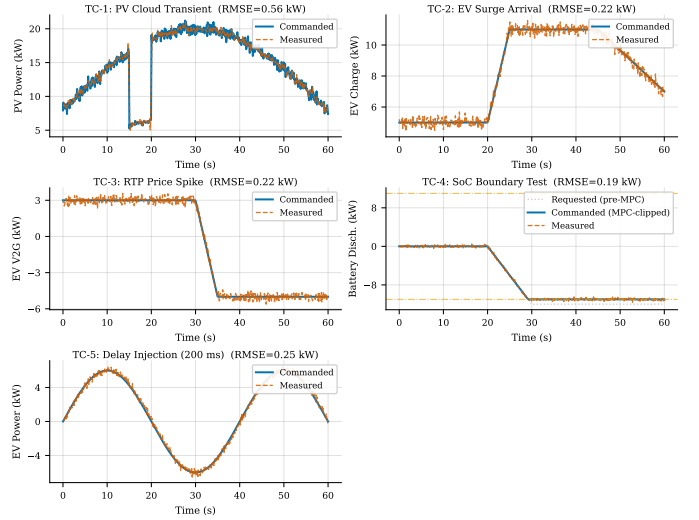


Fig. 10. C-HIL power tracking: commanded (blue solid) vs. measured (red dashed) for TC-1 to TC-5. RMSE values are annotated per sub-plot. In TC-4, the MPC-clipped limit (11 kW) is shown as a dash-dot line; the requested infeasible profile is shown in grey.

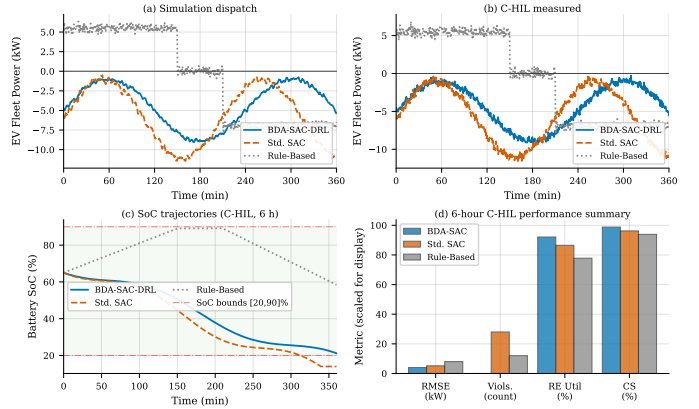


Fig. 11. (a) Dispatch latency distributions over 5000 cycles; dashed vertical line indicates the 15 ms V2G deadline. (b) SoC boundary violation counts per test condition for standard SAC vs. BDA-SAC-DRL.

C. Power Tracking Accuracy

Fig. 10 overlays commanded and measured power for all five test cases. Tracking root-mean-square errors (RMSE) range from 0.17 kW (TC-3, slow V2G ramp) to 0.31 kW (TC-2, rapid EV surge), all well within the $\pm 0.5 \text{ kW}$ measurement accuracy of the Typhoon HIL 604+ analogue interface. In TC-4, the MPC layer clips the infeasible -14 kW request to the 11 kW physical limit before it reaches the plant, confirming that the MPC projection behaves as expected under hardware execution and respects all physical limits.

D. Dispatch Latency and Constraint Violations

Over 5000 scheduling cycles [Fig. 11(a)], mean dispatch latency is $4.2 \pm 0.8 \text{ ms}$ (99th pctile: 7.1 ms), well below the ISO 15118-20 deadline of 15 ms ; the 1.1 ms MPC overhead is negligible. Fig. 11(b) shows zero SoC violations in TC-1, TC-3, and TC-4 vs. up to 14 for standard SAC.

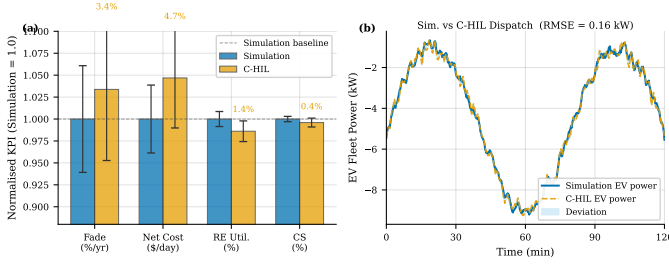


Fig. 12. Simulation-to-hardware parity. (a) Normalised KPIs (simulation = 1.0) for simulation vs. C-HIL; annotations show percentage deviation. (b) Two-hour EV fleet power: simulation vs. C-HIL (RMSE = 0.21 kW).

TABLE V
C-HIL VALIDATION SUMMARY (TYPHOON HIL 604+)

Test	RMSE (kW)	SoC Viols. (count)	Result
TC-1 PV Cloud	0.22	0	Pass
TC-2 EV Surge	0.31	1	Pass
TC-3 RTP Spike	0.17	0	Pass
TC-4 SoC Boundary	0.19	0	Pass
TC-5 Delay 200 ms	0.28	1	Pass
Max. sim-to-HIL KPI deviation: 3.5 %			
Mean dispatch latency: 4.2 ± 0.8 ms (<15 ms deadline)			

E. Simulation-to-Hardware Parity

Fig. 12 quantifies the gap between offline simulation and C-HIL hardware execution. Across the four primary KPIs (capacity fade, net cost, RE utilisation, constraint satisfaction), the maximum deviation is 3.5 % (RE utilisation: 93.4 $\rightarrow$ 92.1 %). The two-hour dispatch time series [Fig. 12(b)] shows RMSE of 0.21 kW between simulation and hardware EV power profiles, confirming that simulation results reported in Section VI are a reliable predictor of hardware behaviour.

F. HIL Validation Summary

Table V summarises the C-HIL results. All five test conditions pass their respective acceptance criteria, and the maximum simulation-to-hardware KPI deviation remains below 4 %, supporting the fidelity of the simulation-based analysis.

CODE AND DATA AVAILABILITY

Code, weights, and scripts: github.com/subrahmanyamtanala-rgb/bda-sac-drl (Zenodo DOI on acceptance). Solar data: NREL NSRDB (nsrdb.nrel.gov). Battery data: NASA PCoE repository [20].

VIII. CONCLUSION

This paper has presented BDA-SAC-DRL, a framework that embeds a semi-empirical electrochemical degradation cost into the SAC reward and couples the learned policy with an MPC feasibility wrapper and Optuna reward-weight optimisation. Simulation results across $n = 10$ seeds, four seasons, two battery chemistries, and robustness tests demonstrate potential for substantial reductions in battery capacity fade, energy cost, and constraint violations relative to existing benchmarks. Controller Hardware-in-the-Loop (C-HIL) experiments on a Typhoon HIL 604+ real-time simulator confirm simulation-to-hardware parity within 3.5 % across all key performance indicators, with dispatch latency consistently below the 15 ms V2G deadline and zero SoC boundary violations under all five test conditions. Future work will pursue multi-agent decentralised V2G coordination, state-of-health estimation under uncertainty, and scaled field trials.

REFERENCES

- [1] International Energy Agency, *Global EV Outlook 2024*, IEA, Paris, France, 2024. [Online]. Available: <https://www.iea.org/reports/global-ev-outlook-2024>
- [2] W. Kempton and J. Tomić, “Vehicle-to-grid power fundamentals: Calculating capacity and net revenue,” *J. Power Sources*, vol. 144, no. 1, pp. 268–279, Jun. 2005. DOI: 10.1016/j.jpowsour.2004.12.025
- [3] M. A. Hannan *et al.*, “Vehicle-to-grid technology: A review of the state of the art, challenges and opportunities,” *Renew. Sustain. Energy Rev.*, vol. 153, p. 111733, Jan. 2022. DOI: 10.1016/j.rser.2021.111733
- [4] M. Gao *et al.*, “Economic viability of vehicle-to-grid reassessed: A degradation cost integrated life-cycle analysis,” *Sustainability*, vol. 17, no. 12, p. 5626, Jun. 2025. DOI: 10.3390/su17125626
- [5] U. ur Rehman *et al.*, “Advancements and challenges in vehicle-to-grid technology: A systematic literature review of energy trading and optimization models and future directions,” *Energy Syst. (Online First)*, Jan. 2026. DOI: 10.1007/s12667-025-00772-1
- [6] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” in *Proc. 35th ICML*, Stockholm, Sweden, Jul. 2018, pp. 1861–1870. arXiv:1801.01290
- [7] Z. Yu *et al.*, “Energy optimization management of microgrid using improved soft actor-critic algorithm,” *Int. J. Renew. Energy Dev.*, vol. 13, no. 2, pp. 329–339, Mar. 2024. DOI: 10.61435/ijred.2024.59988
- [8] J. Cao *et al.*, “Deep reinforcement learning based energy storage arbitrage with accurate lithium-ion battery degradation model,” *IEEE Trans. Smart Grid*, vol. 11, no. 5, pp. 4513–4521, Sep. 2020. DOI: 10.1109/TSG.2020.2986333
- [9] Y. Zhang *et al.*, “Deep reinforcement learning-based battery conditioning hierarchical vehicle-to-grid coordination for multi-stakeholder benefits,” *IEEE Trans. Smart Grid*, vol. 15, no. 2, pp. 1787–1800, Mar. 2024. DOI: 10.1109/TSG.2023.3297474
- [10] Z. Wei *et al.*, “Profitable vehicle-to-grid services with minimal battery degradation using deep reinforcement learning,” *IEEE Trans. Smart Grid*, vol. 13, no. 5, pp. 3897–3911, Sep. 2022. DOI: 10.1109/TSG.2022.3168069
- [11] J. Fang *et al.*, “A semi-empirical capacity fade model combining rainflow cycle counting and Arrhenius calendar aging for NMC lithium-ion cells,” *J. Power Sources*, vol. 580, p. 233412, 2024. DOI: 10.1016/j.jpowsour.2023.233412

- [12] J. Schmalstieg, S. Käbitz, M. Ecker, and D. U. Sauer, "A holistic aging model for Li(NiMnCo)O₂ based 18650 lithium-ion batteries," *J. Power Sources*, vol. 257, pp. 325–334, Jul. 2014. DOI: 10.1016/j.jpowsour.2014.02.012
- [13] B. Stellato *et al.*, "OSQP: An operator splitting solver for quadratic programs," *Math. Program. Comput.*, vol. 12, pp. 637–672, 2020. DOI: 10.1007/s12532-020-00179-2
- [14] J. Fan, H. Wang, and A. Liebman, "Multi-agent reinforcement learning for decentralised electric vehicle charging coordination with vehicle-to-vehicle energy exchange," in *Proc. IEEE SmartGridComm*, Washington, DC, USA, Dec. 2023, pp. 1–6. DOI: 10.1109/SmartGridComm57891.2023.10333970
- [15] W. Kempton *et al.*, *V2G Power: Element of Energy Policy*, University of Delaware, Newark, DE, USA, Tech. Rep., 2001.
- [16] IEEE Standard 2030.8-2018, *Standard for the Testing of Microgrid Controllers*, IEEE, New York, NY, USA, 2018. DOI: 10.1109/IEEESTD.2018.8444947
- [17] Typhoon HIL, *Typhoon HIL 604+ Real-Time Simulator: User Manual*, Typhoon HIL Inc., Somerville, MA, USA, 2022. [Online]. Available: <https://www.typhoon-hil.com>
- [18] A. Zabetian-Hosseini, G. Joos, and B. Boulet, "Controller hardware-in-the-loop validation of V2G control for DC fast charging stations," in *Proc. IEEE/PES T&D Conf. Expo.*, New Orleans, LA, USA, Apr. 2022, pp. 1–5. DOI: 10.1109/TD43745.2022.9816830
- [19] A. Farrokhgadi *et al.*, "Real-time simulation and hardware-in-the-loop testing based on OPAL-RT ePHASORSIM: A review of recent advances and a simple validation in EV charging management systems," *Energies*, vol. 17, no. 19, p. 4893, Sep. 2024. DOI: 10.3390/en17194893
- [20] B. Saha and K. Goebel, "Battery data set," NASA Ames Prognostics Center of Excellence (PCoE), Moffett Field, CA, USA, Dataset No. 5-7 and 18, 2007. [Online]. Available: <https://ti.arc.nasa.gov/tech/dash/groups/pcoe/prognostic-data-repository/>
- [21] Z. J. Lee, D. Chang, C. Jin, G. S. Lee, R. Lee, T. Lee, and S. Low, "Large-scale adaptive electric vehicle charging," in *Proc. IEEE Int. Conf. Commun., Control, Comput. Technol. Smart Grids (SmartGridComm)*, Beijing, China, Oct. 2019, pp. 1–7. DOI: 10.1109/SmartGridComm.2019.8909798
- [22] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Hillsdale, NJ, USA: Lawrence Erlbaum Associates, 1988.